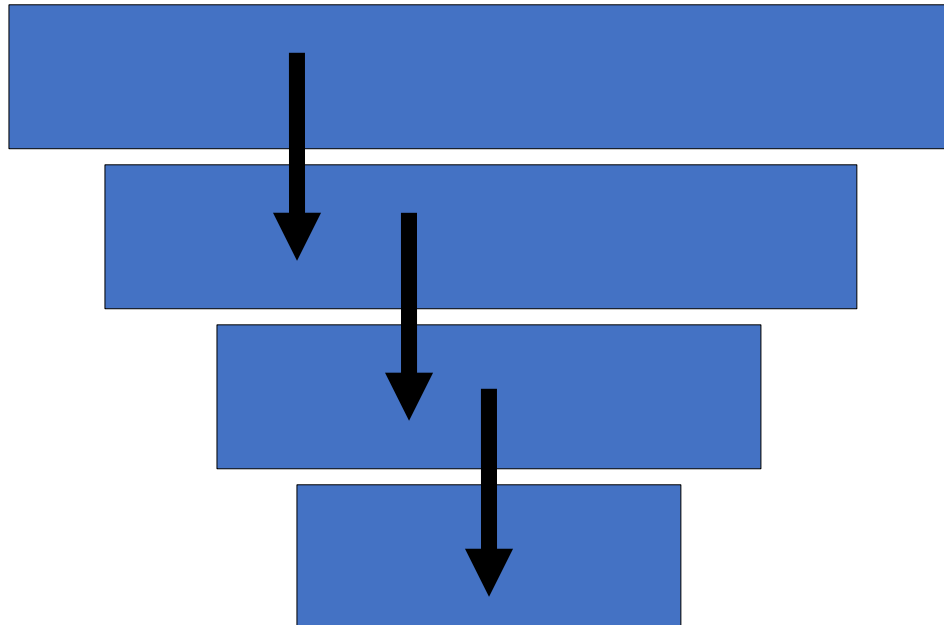


The role of the Introduction

Fish 290

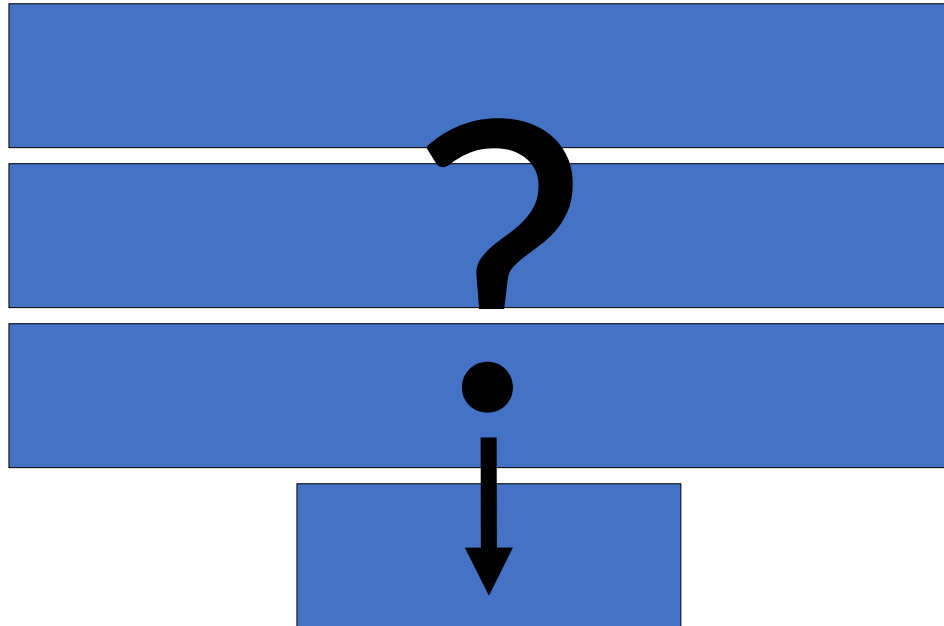
From Tuesday: **The Introduction**

- **Begin with the broadest scope** and get progressively narrower, leading steadily to the **statement of objectives** in the last paragraph of the Introduction.



From Tuesday: **The Introduction**

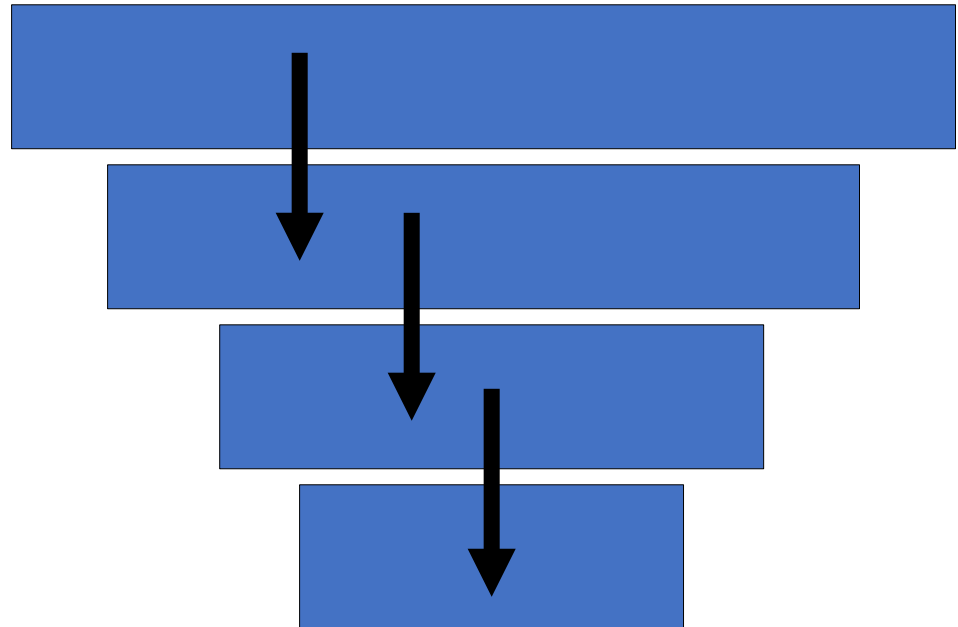
- **An Introduction **DON'T**:** give a literature review on everything you know about the subject, then tack on your project goals at the end.



From Tuesday: **The Introduction**

- **Begin with the broadest scope** and get progressively narrower, leading steadily to the **statement of objectives** in the last paragraph of the Introduction.

Today: how do we go from broad to narrow gracefully?



The Introduction: 3 components

1. Define a research territory
2. Establish a niche within that territory
3. Occupy that niche

The Introduction: 3 components

1. Define a research territory

- Begins with the first sentence of your Introduction!
- The role of the first paragraph (*one of the 5 pivotal paragraphs*) is to **set boundaries** on the broadest scope of your paper.
- These boundaries depend on your journal and intended audience.
 - If we are writing for a fisheries journal, we may start with fish or management concepts or themes
 - If we are writing for an ecology/behavior/evolution journal we may start with general themes across taxa before narrowing down to fish
- By doing this, you can later explain why your study is important in this broad context

The Introduction: 3 components

2. Establish a niche within the research territory

- Next you **point out a gap** in the research territory you established in the first paragraph or two
- Why are we doing this study? To learn something we didn't know before!
- Lots of things have never been done but aren't worth doing because of time or money. Lots of things aren't known but aren't important to know either. You need to explain **why it is important for our broader understanding** to fill this gap you've identified.
- The best examples of this build tension or captures the reader's curiosity, makes the reader *want* to know the solution

The Introduction: 3 components

3. Occupy the niche

- Here is where you provide the solution the reader is waiting for!
- Define your research question, hypotheses, and overall approach. The more specific, the better.
- Also need to explain **why your study fills the gap you just identified**. It may be obvious to you but help the reader out! (we're tired)
- Sometimes the last sentence of the Introduction gives the key finding.
- Includes *one of the five pivotal paragraphs!* Why so important?

A note on structure within paragraphs

Each paragraph should have a **topic sentence**, a series of sentences in logical order, and a **concluding sentence** that provides a link to the next paragraph.

Awkward transitions within a paragraph, and between paragraphs are a sure way to lose readers. Remember you want them to follow you from broad to narrow without losing them along the way!

Middle sentences should **provide references** that support and/or elaborate on the topic sentence. If you find yourself deviating from the topic sentence, either rewrite it or **start a new paragraph**.

Class/Homework Activity #1

1. Reverse Outline Activity

- One popular and help way to write a paper is to start with an outline, where you use bullet points, numbers, or ideas to block out the structure of a paper.

- This method works in reverse if you want to critically analyze the structure of a scientific paper

- Read the introduction of the paper by Orejas et al. (2016) and create an outline showing their thought process and how they guide us through the background introduction towards their objectives. On the PDF document, highlight any topic sentences or transitional sentences you can identify.

- Turn in both your highlighted PDF and your outline on Canvas by Wednesday

Class/Homework activity #2

1. Read the “mock introductions to edit” document which contains intros written by students from past papers
2. Read the instructions in that document, edit it, save it, and submit it on canvas.

Class/Homework activity #3

Find data for your paper hypothesis.

What kind of data do I need?

- Control
- **Variables** (independent, dependent)
- Independent **replicates**

Where can I find data?

- Google search for databases
- Gather from several published studies that measure the same type of variables
- UW libraries data resource page